

Weekly Progress Report

Week 04 — Data Science & Machine Learning Internship

STUDENT NAME

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DOMAIN

Data Science and Machine Learning

SELECTED PROJECTS

- **Agriculture:** Prediction of Agriculture Crop Production in India
- **Smart City:** Forecasting of Smart City Traffic Patterns

1. Tasks & Progress

• Milestones Achieved:

- Completed theoretical modules on Machine Learning foundations, supervised learning systems, regression techniques, density estimation, and optimization theory.
- Studied key algorithms including Naive Bayes, K-Nearest Neighbors, Perceptron, K-Means, Logistic Regression, Linear Models (SVMs), and Decision Trees / Random Forests.
- Attempted and evaluated weekly assessment covering Naive Bayes posterior estimation, feature normalization techniques, Decision Trees (scikit-learn implementation), and Random Forest performance metrics.

• Key Work Completed:

- **Agriculture Project:** Implemented Supervised Regression models to predict crop yield based on environmental features, and applied Maximum Likelihood Estimation (MLE) and Parzen Windows for density estimation of climate parameters.
- **Smart City Traffic Project:** Implemented Time-Series Regression and Gradient Descent optimization for traffic forecasting, and explored Support Vector Regression (SVR) and Kernel methods for non-linear traffic flow modeling.

2. Challenges & Hurdles

- **Obstacles Encountered:**

- Handling high variance and bias trade-offs during initial regression model fitting on complex agricultural datasets.
- Convergence issues and hyperparameter tuning difficulties in convex optimization algorithms (Gradient Descent vs Stochastic Gradient Descent) for real-time traffic flow prediction.
- Managing feature magnitude disparities across heterogeneous datasets requiring normalization.

- **Solutions Applied:**

- Applied Regularization techniques (Ridge/Lasso) and Support Vector Machines with Kernel Tricks to manage non-linearity and overfitting.
- Utilized learning rate scheduling, feature scaling/normalization, and Stochastic Gradient Descent (SGD) for faster convergence in optimization tasks.

3. Lessons Learned

- **ML Foundations & Models:** Mastered fundamental models including Naive Bayes, KNN, Perceptron, K-Means, Decision Trees, and evaluation metrics (Accuracy, F1-score, MSE).
- **Density Estimation & Normalization:** Learned Maximum Likelihood Estimation, Bayesian posteriors, and the critical role of data normalization in distance-based algorithms and gradient descent convergence.
- **Optimization & Linear Models:** Gained theoretical and practical understanding of smooth convex minimization (Gradient Descent) and large-margin classifiers (SVMs, SVR, Kernel methods).